

Example Problem

Calculate the tensile stress that is applied along the $[1\bar{2}0]$ axis of a gold crystal to cause slip on the $(1\bar{1}\bar{1})[0\bar{1}1]$ slip system. The critical resolved shear stress is 10 MPa.

The angle between the tensile axis $[1\bar{2}0]$ and the normal to the slip plane $(1\bar{1}\bar{1})$ is:

$$\cos \phi = \frac{(1)(1) + (-2)(-1) + (0)(-1)}{\sqrt{(1)^2 + (-2)^2 + (0)^2} \sqrt{(1)^2 + (-1)^2 + (-1)^2}} = \frac{3}{\sqrt{5}\sqrt{3}} = \frac{3}{\sqrt{15}}$$

The angle between the tensile axis $[1\bar{2}0]$ and the slip direction $[0\bar{1}1]$ is:

$$\cos \lambda = \frac{(1)(0) + (-2)(-1) + (0)(1)}{\sqrt{(1)^2 + (-2)^2 + (0)^2} \sqrt{(0)^2 + (-1)^2 + (-1)^2}} = \frac{2}{\sqrt{5}\sqrt{2}} = \frac{2}{\sqrt{10}}$$

Since CRSS = 10 MPa,

$$\sigma = \frac{P}{A} = \frac{\tau_{CRSS}}{\cos \phi \cos \lambda} = \frac{10}{\left(\frac{3}{\sqrt{15}}\right)\left(\frac{2}{\sqrt{10}}\right)} = \boxed{8.16 \text{ MPa}}$$